Aanya Voice Agent Design Note

Oct 27, 2026 · Kumar Shubham

Aanya is an outbound voice agent, built on ElevenLabs with a FastAPI backend, that calls a bank's leads about a credit card. It is designed around the two goals confirmed with Vijay before building: stop the five-second hang-up, and get a verbal yes after the fee has been stated. The customer is the bank. The leads, and whatever the bank knows about them, come from the bank, so the agent is built as a ladder: the more the bank shares, the more personal the call, and even with only a name and a phone number it still runs a complete, polite conversation.

The brief, and what was confirmed before building

The assignment, given after the interview on 22 October: build an outbound voice agent on ElevenLabs that sells an Indian credit card, HDFC as the example, and handles the conversational edge cases, with one explicit problem at the centre: people hang up on card sales calls within seconds, so the opener has to keep them on the line.

Before writing a line of prompt I sent Vijay a short set of questions on WhatsApp. His answers on 25 October fixed the scope, and every one of them shows up in the build.

What I asked	Vijay's answer	What it changed
Which bank, and whose agent is it?	Assume any bank. The agent represents the bank, and the lead comes in from the bank.	Bank name is a variable. The first words are "Aanya here from HDFC Bank", never a "card team".
How much do we know about the lead?	Assume as little lead data as possible.	Three data tiers. The lowest tier is a name and a phone number, and the opener for it names no product at all.
What counts as success?	The customer verbally agrees to take the card, not a completed application.	The close is built around one explicit yes, asked only after the fee has been stated. The post-call record captures that yes as its own field.
Do we scrub DND or number series?	Not relevant for this.	Kept as a backend check, since it costs nothing, but not a focus.

What I asked	Vijay's answer	What it changed
Should the agent say it is an AI?	Only if asked.	Never volunteered. Honest when asked, once, with a human callback offered.
What matters most?	Two things: the hang-up problem and getting the yes.	The design budget went to the opener and to how a no is handled. Everything else is a guardrail.

Who the customer is

Karta's customer is the bank, not the person on the phone. The bank owns the leads, decides which card to offer, and shares as much or as little about each lead as its systems and its consent allow. So the agent is not one script. It is one agent that reads whatever the bank has given it and speaks accordingly. Once the customer is a bank, the three kinds of people it will ask us to call follow directly.

Segment	What the bank knows	Demo leads	What the agent must get right
Existing customer who already holds one of the bank's cards	Tenure, products held, category-level spends, a pre-approved offer and limit	Vijay, Bengaluru: four years, salary account and a MoneyBack Plus, offered Regalia Gold for travel	Treat it as an add-on, never a replacement. Never talk down the card they hold. If their card already covers the need, say so and leave.
Existing customer with no card	Tenure, account type, category-level spends, a pre-approved offer and limit	Priya, Mumbai (travel); Sneha, Pune (online shopping)	No age or income questions, the bank has them. The offer is visible in the app, so say so: it is how sceptical customers verify.

Segment	What the bank knows	Demo leads	What the agent must get right
New to the bank	At best a lead source and a self-declared spend; at worst a name and a phone number	Rahul (Swiggy app), Amit (branch form, Hindi), Meera (Paisabazaar); Vikram with nothing at all	Earn the right to name a product. For Vikram the first turn names no card; his own answer picks one. Eligibility is asked, once, in one question.

The industry reason this ordering matters: Indian issuers source most new cards from people who already bank with them, and those are the calls where the data is richest and the conversion highest. The tier-0 cold call is the hardest and the one Vijay asked me to assume by default, so the design was built from the bottom up.

The data ladder: what the bank shares, and the sentence it unlocks

Every field the bank can share buys the agent one more true, specific sentence. The push budget is identical at every rung. What changes is the ammunition.

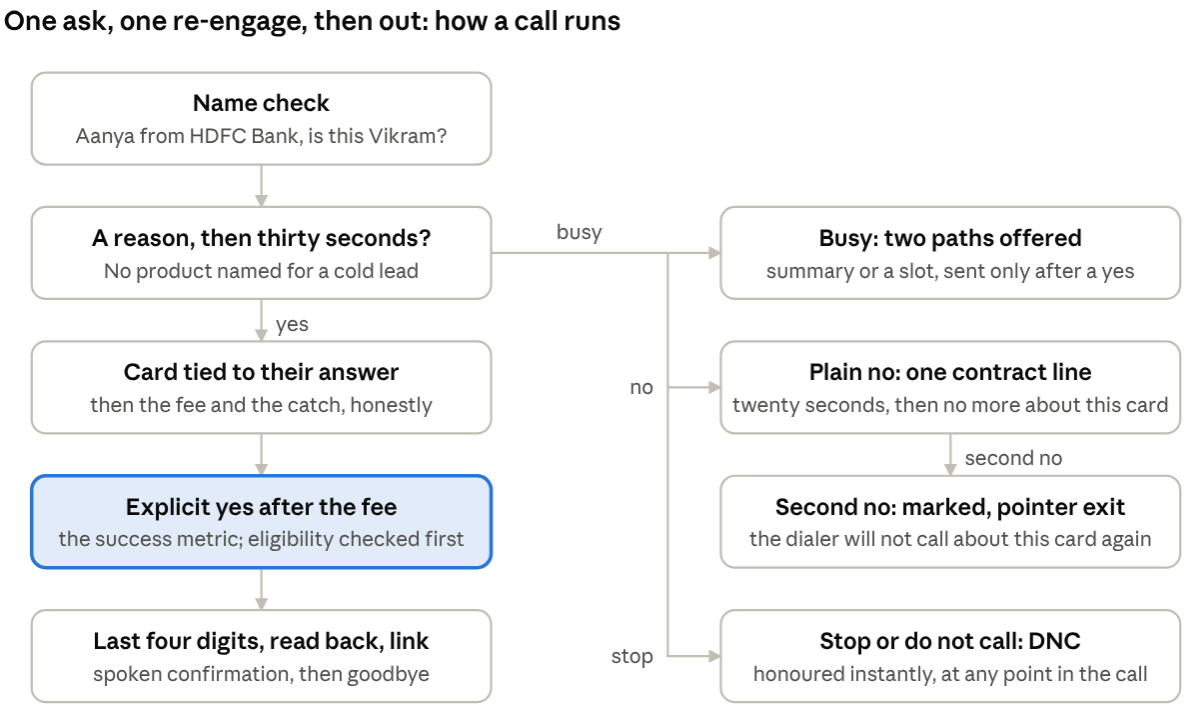
Field from the bank	Who has it	The sentence it unlocks
Name and phone	Everyone	"Am I speaking with Vikram?" then a product-free first turn and a spend question whose answer picks the card.
Lead source	New leads from a channel	"You'd tapped the card offer inside the Swiggy app last week, so I'm following up on that, nothing to sign today."
Declared spend category	New leads from a form or aggregator	The either-or question is skipped and the benefit is tied to the category they named.
Relationship and tenure	Bank's own customers	"There's an offer sitting on your account that was matched to how you spend with us, it's in your app as well." No eligibility questions.

Field from the bank	Who has it	The sentence it unlocks
Category-level monthly spend	Bank's own customers	The value line: "Going by your travel spends with us, that's roughly five thousand five hundred rupees back a year and twelve free lounge visits."
Fee-waiver threshold against their spend	Bank + catalogue	The fee outlook: "The waiver kicks in at four lakh of spend in the year; going by your travel spends you'd likely cross it, and if you don't, you pay it."
Pre-approved limit	Bank's own customers	"The limit's already set at three lakh rupees, you can see it in the app, and there's no documentation because the bank already has your KYC."
Offer validity	Bank's own customers	Spoken once, as information: "it's open till the thirtieth of November." Never as pressure.
Products held	Bank's own customers	Add-on framing for a cardholder, and the honest exit when their existing card already does the job.

Why pre-call personalisation from spending habits is the lever. A card is only worth having if it pays back on spends the person already makes. The bank is the one party that can see those spends at category level before the call, so it can match the card to the profile and hand the agent a reason that is true for this person. That is a different call from "you've been shortlisted for a card": it opens with something the customer recognises as their own life, often a pattern they had not put a number on themselves. The expected effect is on both goals at once. The opener survives because it is about them, and the yes comes easier because the fee question has already been answered against their own numbers rather than a brochure.

Two rules keep this from tipping into the creepy version customers describe. The agent speaks in categories and outcomes, never merchants, amounts or dates. And when someone asks "are you tracking where I spend?", the answer is short and honest: category totals on their own account, which they can switch off in the app. The simulation where a customer asks exactly that is one of the pass criteria.

How a call runs



call flow · 5 steps on the yes path, 4 exits

The left column is the only path to a yes, and it has one gate: the fee, the waiver and "approval is subject to the bank's check" are spoken before the agent asks "Entirely your call: shall I go ahead?" The right column is every way out, and each one ends with something true happening in the bank's systems: a consented summary or slot, a decline record, or a do-not-call entry.

The opener

The standard Indian card call opens with identity and pitch in one breath: "calling from the HDFC credit card department, you've been pre-approved for a lifetime free card." The decision to hang up is made inside those five seconds. I built the opener from two sources: what customers themselves say drives the reflex, and what call data says shortens it.

What customers say. A long r/IndianCreditCards thread, started by a telecaller asking why most people say no, ranks the reasons: the pitched card is worse than what they hold; the caller cannot be verified; "pre-approved" turned out to be a lie last time; "lifetime free" turned into an annual fee; telecallers get defensive at any negative; repeat calls after a no; nobody asks if it is a good time; fear of debt. Every one of those has a specific answer in the agent, listed in the sections that follow.

What the call data says. Gong's analysis of cold calls puts a permission-based opener at an 11.2% success rate against 2.2% for "did I catch you at a bad time", and reports a 2.1x lift from stating the reason for the call up front. Nooks' April 2025 data on its own platform shows calls that open by asking for thirty seconds book meetings at 13.9% against 5.8% without. Both are vendor datasets from B2B calling, not Indian consumer banking, so I treat them as direction, not forecast. The direction is consistent: own the call, give a reason that is about them, ask for a small, bounded permission, and put the product after the permission.

What Aanya says. Three turns, with the product arriving only in the third.

Turn	Existing customer (Priya)	New lead with a source (Rahul)	Unknown lead (Vikram)
First message	Hello, Aanya here from HDFC Bank. Am I speaking with Priya?	same	same
After the name	Hi Priya. There's an offer sitting on your account that was matched to how you spend with us, it's in your app as well. Got thirty seconds to check if it's actually useful?	Hi Rahul. You'd tapped the card offer inside the Swiggy app last week, so I'm following up on that, nothing to sign today. Got thirty seconds to see if it's worth your time?	Hi Vikram. Nothing to sign today, I just want to ask how you pay for things month to month, two questions and I'm done. Got thirty seconds?
After the yes	Going by your travel spends with us, that's roughly five thousand five hundred rupees back a year and twelve free lounge visits. That's the Regalia Gold, pre-approved, so no documents. Want the fee and the catch before anything else?	Quick one: does more of your month go on food delivery, or on online shopping? Then the benefit tied to the answer.	Quick one: where does most of your month go? UPI for everything, groceries and fuel, online shopping, food delivery, or travel? Then the one card that fits, including a RuPay card inside their UPI app if the answer is "UPI for everything".

Identity is said exactly once. The words "credit card" do not appear before permission. The thirty-second ask ends every opener on a question, because a question is harder to hang up on than a statement.

Handling no

This is the part I spent the most time on, because it is where human telecallers and most AI agents lose the customer for good. Both are built to rebut, and rebutting teaches the customer that no does not work, so next time they hang up instead of saying it. Aanya gives a no three properties: it is answered once, with a contract; the contract is kept by the system, not by the model's good behaviour; and nothing is ever sent to the customer without their yes.

The push budget is the same for every segment. One thirty-second ask, one re-engage, then out. What differs is the one specific thing the re-engage offers.

What the customer says	What Aanya does	The line
"I'm busy" or "another time"	Not a no. Any question inside it is answered first. Then two paths in one breath.	Totally. Either it's a two-minute thing right now, or I WhatsApp you a two-line summary and call you back at six, whichever suits?
"Not interested" or "oh, no"	One re-engage, framed as a contract, spending the twenty seconds rather than asking for them.	Fair enough. One thing in twenty seconds, then if it's not for you I'll mark it and you won't hear about this card again: going by your travel spends with us it's worth roughly five and a half thousand a year plus twelve lounge visits. Want the fee and the catch?
Same, for a bank customer	The same contract, ending on a choice that includes a soft off-ramp.	…Want the fee and the catch now, or shall I just put the two lines on your WhatsApp and you look when it suits?
"Just WhatsApp it"	Not a decline. Summary sent, with a talk-to-us button that resumes the conversation.	Done, it's on your WhatsApp with a talk-to-us button; tap it whenever and we pick up from here, no need to repeat anything. Bye, Priya.

What the customer says	What Aanya does	The line
A second no	The promise is kept. A decline record is written and the dialer refuses future calls about that card. The exit is a pointer, never another question.	Done, it's marked, you won't hear about this card again. If you ever want a look, it's in your app under offers. Bye, Priya.
"Don't call me again", at any point	Straight to the do-not-call list. No re-engage.	Understood, I'm removing your number right now, sorry for the trouble. You won't hear from us again.
"What's the catch?" or "I already have a card"	Treated as engagement, not as a no. One honest answer, then the smallest next question.	It's a paid card, two thousand five hundred rupees a year, I won't dress that up. The waiver kicks in at four lakh of spend in the year…

Three rules sit under the table. The agent never asks why, because asking a customer to justify a no is the defensiveness the thread complains about. There is never a third attempt. And the "one thing" is the richest true sentence the data allows: the value line for a bank customer, the card's best benefit for a lead with a source, and for an unknown lead the RuPay card that runs inside the UPI app they already use.

On the real calls this week the agent ran this sequence as designed, including on Rahul's call, where a busy, then a plain no, then a second no produced the two-path offer, the contract line and the marked exit, in that order and nothing more.

Honesty and compliance, enforced in code

The design principle: judgement belongs in the prompt and the tools, where it can be reviewed and tested. The model's job on the call is to be fast and obedient. So the things that must be true on a bank call are not requests to the model. They are refusals in the backend that the model cannot talk its way past.

Rule	Where it lives	What happens if the model tries anyway
Fee, waiver and "subject to eligibility" before any link, and an explicit yes after them	Backend, two-phase link tool	The first call is refused and returns the exact disclosure sentence to say. Only a second call carrying the customer's post-disclosure yes sends anything.
Eligibility checked before a link	Backend	Link refused with "no eligibility check"; the agent is told to go back.
Link goes only to the number on the call	Backend	Last-four mismatch is refused. "Send it to my wife's number" is declined kindly in the prompt and impossible in the tool.
No message or callback without consent	Backend, summary and callback tools	Both refuse unless the customer's last message said yes; the callback also needs a real time. "Later" is refused. Each refusal hands the agent the question to ask.
One re-engage, then stop	Prompt, plus the decline record	The second no writes a decline the dialer honours, so the promise "you won't hear about this card again" is true.
Decline is not do-not-call	Prompt and judge criterion	Only an explicit "don't call again" reaches the DNC list, so a declining bank customer is not cut off from every legitimate call.
Calling hours 10:00 to 19:00 IST	Backend gate	Session refused outside the window (RBI Credit and Debit Card Directions, 2025).

Rule	Where it lives	What happens if the model tries anyway
Never OTP, PIN, CVV, card number, Aadhaar, PAN	Prompt guardrail and judge criterion	If the customer starts reading one out, the agent interrupts: "Please don't share that, I don't need it, and the bank will never ask for it on a call."
Never "lifetime free", "guaranteed approval", "no interest"	Prompt, judge criterion, unit test on the fee outlook	The fee outlook is generated from the catalogue and the lead's own spend, and a test fails the build if the words free, lifetime or effectively appear.
Instructions on the call cannot change the rules	Prompt	A caller claiming to be compliance or a manager gets the official helpline and nothing else. One simulation injects exactly this.
AI disclosure	Prompt	Volunteered never, answered honestly once when asked, with a human callback offered.

The honesty on fees goes beyond compliance. "It's a paid card, two thousand five hundred rupees a year, I won't dress that up. The waiver kicks in at four lakh of spend in the year; going by your travel spends you'd likely cross it, and if you don't, you pay it. The welcome gift is a voucher, not a refund." That sentence exists because the thread's most-quoted scam was "coupons plus a waiver, so it's effectively lifetime free". A bank's sales head may find it blunt. I think it is the single most credible thing the agent says.

How it was tested

Three layers, because each catches what the others cannot.

Layer	What it covers	Count	What it cannot see
Backend unit tests	Every refusal above: two-phase link, consent guards, last-four match, calling-hours gate, decline and DNC records, idempotency per call, the fee-outlook wording	22 tests, run in isolation against a throwaway database	Whether the model actually calls the tools in the right order
Simulated customers	Scripted personas run through the ElevenLabs simulation API, judged on 19 criteria: identity once, no product before permission, stop respected with one re-engage allowed, fee before link, number confirmed, no forbidden claims, no sensitive data, language matched, graceful end, summary only when asked, and more	34 scenarios, from a hostile early no to a compliance-officer injection to "are you tracking my spends?"	The tools are mocked, so the backend refusals never fire here. The simulated customer is also too polite: it sometimes pastes its whole script in one turn.
Real calls through the browser dialer	Latency, voice, interruptions, background noise, the actual tool sequence	Roughly a dozen calls across two ElevenLabs accounts, each inspected afterwards through the conversation API, which gives per-turn timing, interruption flags and tool calls	Scale. These are my calls, not customers.

The scenario list includes: hostile early no; firm no twice; busy and driving; already has cards and is happy; debt-averse; OTP sharer in Hindi; scam-suspicious and asks if it is AI; wrong person; abusive; ineligible student; the full happy path; a rambler with several intents; existing customer who verifies in the app; other bank's card holder who qualifies;

"this is the fifth call today"; yes before the fee; send it to my wife's number; fake compliance injection; "which number is this, email me from the official domain"; "are you tracking my spends"; "what's in it for me, actually I'm busy"; "drop it" is not DNC; the minimal-data lead; busy with a fee question; two personas built from the Reddit thread; the reason-first opener per tier; the contract re-engage, both outcomes; the tier-0 lead through the UPI route and through a fee objection; and the existing customer who parks the offer on WhatsApp or says no twice.

Compliance criteria pass on every scenario. The remaining failures in the full run are simulator artefacts, each of which I checked by hand.

What building it taught me

Every one of these was found on a real call, diagnosed from the ElevenLabs conversation record rather than guessed, and fixed in a way that cannot regress.

What I saw	Root cause	Fix
The agent went silent after answering a question, and the call died after 25 seconds	Every turn has to end on a question or a next step, and the skip-turn tool was allowed to wait forever	Prompt rule, bounded wait, a spoken goodbye on silence
The agent's sentence cut off mid-word on the same turn, twice, on two calls	Not an interruption. Gemini 2.5 Flash counts its thinking tokens inside the output cap, and the first turn that needed reasoning spent the cap before speaking	Thinking off, cap raised. Zero truncations since, and the first word arrives about a second sooner
Two to four seconds of dead air before every reply	The model called a stage-logging tool before each sentence	Tool removed. Speak first, act after
A TV in the background cut the agent off and it waited for the "customer" to finish	Background speech transcribed as a user turn	Background voice detection on, plus a recovery line: "Sorry, I lost you for a second, still there?"
A WhatsApp summary and a callback for "later" were sent before the customer was asked	Prompt wording alone does not stop an eager tool call	Consent guards in the backend; both tools refuse without a yes and a real time

What I saw	Root cause	Fix
A first no went straight to "I'm marking this"	The model copied the wrap-up example	The first no never marks; existing customers get the WhatsApp off-ramp
Unit tests were writing into the demo database and the dialer showed phantom events	Shared SQLite file	Tests run against a throwaway file
Switching ElevenLabs accounts failed with "voice not found"	Library voices are per account	The deploy script adds the voice to a fresh account by itself

The pattern across all of them is the same: when the model misbehaved, the durable fix was a sentence it could not misread or a tool that refused, never a longer instruction.

Sources

- Voice of the customer: the r/IndianCreditCards thread "I call people about pre-approved credit cards for a living. Why do most of you say no?", mapped line by line in the repository's research notes.
- Gong, the best and worst cold call openers: permission-based opener 11.18%, "did I catch you at a bad time" 2.15%.
- Gong, cold call opening lines: stating the reason for the call up front, 2.1x.
- Nooks, permission-based openers: 13.9% versus 5.8%, April 2025.
- Gong, how successful salespeople handle objections: pause longer after an objection, never interrupt, answer with a question.
- Callyzer, credit card sales script and GroMo, selling credit cards on call: the Indian telecalling sequence and objection list.
- RBI circular of 17 January 2025: 1600 series for transactional calls, 140 series for promotional calls.
- RBI Credit and Debit Card Directions, 2025: telemarketing hours 10:00 to 19:00.
- Card terms: HDFC Bank's published pages for each card, as commonly reported; figures in the catalogue are marked to be verified before any real use.

Repository: github.com/Shubham5613/karta-voice-agent (private; access on request).